

Efficient CNN-SWT-Transformer for Fog-Based ECG Denoising

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Abstract—Electrocardiogram (ECG) monitoring is crucial for early detection of cardiovascular diseases, but real-world signals are often contaminated by various factors. Conventional filtering is computationally efficient but limited in suppressing diverse noise types without distorting morphology. Recent deep learning methods achieve superior denoising, yet cloud-based deployment incurs latency, bandwidth, and privacy concerns. In this work, we adapt the CS-TRANS denoising block into a lightweight CNN-SWT-Transformer architecture tailored for fog computing. The proposed model has only 0.15M parameters and runs in real time on a Raspberry Pi 3. On synthetic MIT-BIH + NSTDB data (72k windows, 2s each), it improves signal-to-noise ratio (SNR) by +15.6 dB over a standard band-pass filter, with negligible mean squared error. These results demonstrate that compact deep models can reliably enhance ECG quality on low-cost fog nodes, supporting robust and energy-efficient real-time monitoring.

Keywords—Electrocardiogram, Heart Monitoring, ECG denoising, Stationary Wavelet Transform, Fog Computing

I. INTRODUCTION

Electrocardiogram (ECG) monitoring is an effective tool for early detection of cardiovascular diseases, yet real-world recordings are often corrupted by baseline wander, motion artefacts, and electrode interference, leading to frequent false alarms. Two main strategies have been studied: (i) denoising to suppress noise, and (ii) signal quality indices (SQI) to reject poor signals. Conventional methods such as filtering or wavelet transforms are inadequate, as they cannot remove all noise types and often distort clinically important segments.

Recent deep learning approaches achieve superior denoising by learning nonlinear noise patterns while preserving morphology, but most rely on cloud resources, incurring latency, bandwidth overhead, and privacy risks. Fog

computing addresses this gap by enabling denoising directly at the edge, near the data source. In this work, we adapt the CS-TRANS [1] denoising block into a compact model suitable for deployment on Raspberry Pi 3, demonstrating real-time and energy-efficient fog-based ECG monitoring.

II. METHODOLOGY

A. Fog Node Setup

The fog node was implemented on a Raspberry Pi 3 Model B equipped with a quad-core ARM Cortex-A53 CPU @1.2 GHz and 1 GB RAM. The system ran Raspberry Pi OS (64-bit, Debian 11) with Python 3.11. Lightweight inference was enabled using ONNX Runtime for optimised CPU execution. All denoising modules were exported from PyTorch to ONNX format with opset 13.

B. Dataset Preparation

The MIT-BIH Arrhythmia Database was used as the clean ECG source, while the MIT-BIH Noise Stress Test Database (NSTDB) was used to synthesise noisy ECG windows. Noise types included baseline wander (BW), motion artefact (MA), and electrode motion (EM). Signals were segmented into 2-s windows at 360 Hz with 50% overlap (720 samples, stride = 1 s). Noise was mixed at input SNR levels of {0, 5, 10} dB. Each window was normalised to 0-1. Data were split by patient record into training, validation, and test sets to avoid leakage.

C. Model Architecture

We adapted the denoising module of CS-TRANS architecture into a compact encoder-decoder model (Fig. 1). The encoder has three blocks (Conv1d + ReLU + BatchNorm) with max-pooling, the last with dropout ($p = 0.5$). Then, a learnable CNN-SWT module was applied to extract time-

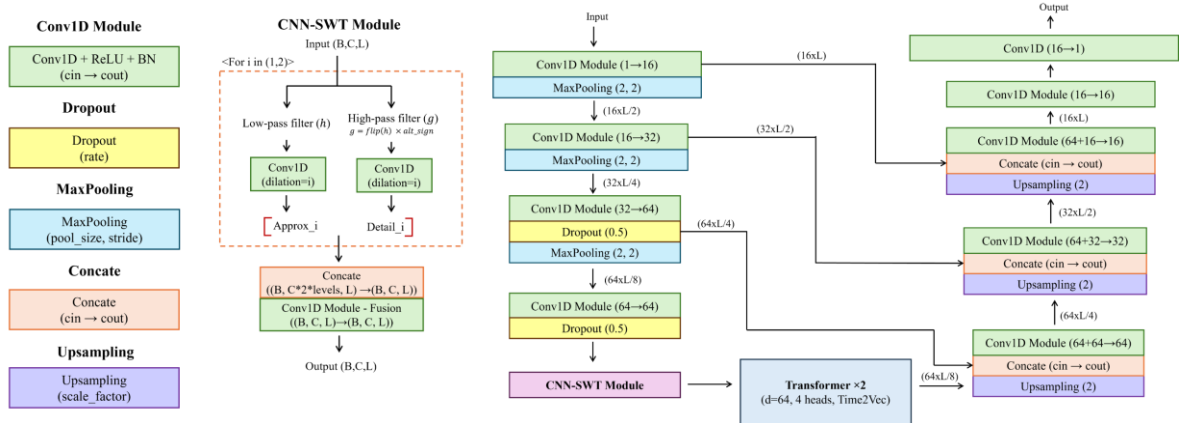


Fig. 1. Overall architecture of the proposed lightweight CNN-SWT-Transformer denoiser.

frequency features. In this module, a low-pass kernel is learned, while its high-pass counterpart is derived by flipping and alternating signs. At multiple dilation scales, approximation and detail coefficients are extracted and fused by a 1×1 projection, enabling efficient representation of both baseline drift and high-frequency artefacts. Compared to the original CS-TRANS, our implementation simplifies the architecture by using a two-level CNN-SWT, instead of a deeper multi-scale decomposition. The bottleneck comprises two Transformer layers (4 heads each) with Time2Vec positional encoding. The decoder mirrors the encoder with three stages of nearest-neighbor upsampling, each concatenated with the corresponding encoder skip and followed by a Conv-BN-ReLU block. A refinement conv and a final 1×1 conv predict the noise estimate, which is subtracted from the input to yield the denoised output.

Training was done for 10 epochs with default-setting Adam (lr = 0.001, batch = 64) and the improved MSE loss function, which adds a peak coefficient to enable the model to learn more information at the peak point.

III. RESULT

A. Quantitative Evaluation.

We compared our model with a standard band-pass + notch filter (0.5–40 Hz, 50 Hz notch), commonly used in ECG monitors. While efficient, the filter often distorts ST segments and fails to suppress motion or electrode artifacts. On the MIT-BIH + NSTDB synthetic test set (72k windows, 2s each), the filter even reduced signal quality. In contrast,

TABLE I. DENOISING PERFORMANCE AT SNR_{IN} = 5 DB

Method	SNR _{in} (dB)	SNR _{out} (dB)	Δ SNR (dB)	MSE
Band-pass Filter	5.00	0.51	-4.49	0.17
Ours	5.00	20.63	+15.63	0.00

TABLE II. DENOISING PERFORMANCE AT SNR_{IN} = 5 DB

Metric	Result
Model size (ONNX)	651 KB
Parameters	0.15M
Mean latency	21.7 ms
Utilization @1 Hz	~2.2% of 1-s cycle

Measured on Raspberry Pi 3 Model B (batch=1, seq_len=720, threads=1). Latency p95 $\approx$ 22.0 ms,

our lightweight DL denoiser improved SNR by +15.6 dB with negligible MSE, showing that even a compact model can effectively handle diverse noise types. These results are summarized in Table I.

B. Qualitative Evaluation.

The noisy signal exhibits baseline wander and motion artifacts that obscure the P–QRS–T morphology. The baseline filter cannot recover these features, while our

denoiser restores morphology close to the clean reference, preserving QRS sharpness and removing both low-frequency drift and high-frequency noise. These visual results are consistent with the quantitative gains reported in Table I.

C. Fog Deployment Metrics.

As summarized in Table II, the proposed model is highly lightweight ($\sim$ 0.15M parameters, 651 KB ONNX) and runs efficiently on a Raspberry Pi 3 Model B. The denoiser achieves inference within only a few tens of milliseconds, with negligible memory overhead. This corresponds to less than 3% CPU utilization for real-time 1-Hz streaming, confirming that fog nodes can host deep learning–based ECG denoisers without compromising latency, energy efficiency, or stability. This shows that even low-cost hardware can reliably support deep learning ECG preprocessing

IV. DISCUSSION AND CONCLUSION.

This study demonstrates that a lightweight CNN–SWT–Transformer denoiser can substantially improve ECG quality compared to conventional filtering, while remaining compact enough for fog-node deployment. Quantitative and qualitative evaluations confirm robust noise suppression with preserved morphology, and edge benchmarks on a Raspberry Pi 3 show that the model runs in real time with minimal resource usage. These findings highlight the feasibility of embedding deep learning–based denoising into healthcare IoT systems to reduce false alarms and support continuous monitoring.

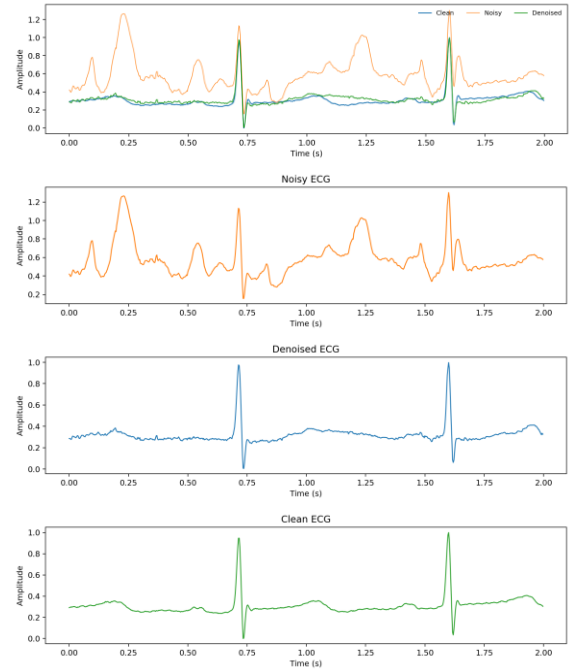


Fig. 2. Example ECG denoising results. Top: overlay of clean, noisy, and denoised signals; middle: noisy and denoised ECG separately; bottom: clean reference ECG.

REFERENCES

- [1] H. Peng, X. Chang, Z. Yao, D. Shi, and Y. Chen, “A deep learning framework for ECG denoising and classification,” *Biomedical Signal Processing and Control*, vol. 94, pp. 106441–106441, May 2024, doi: <https://doi.org/10.1016/j.bspc.2024.106441>.